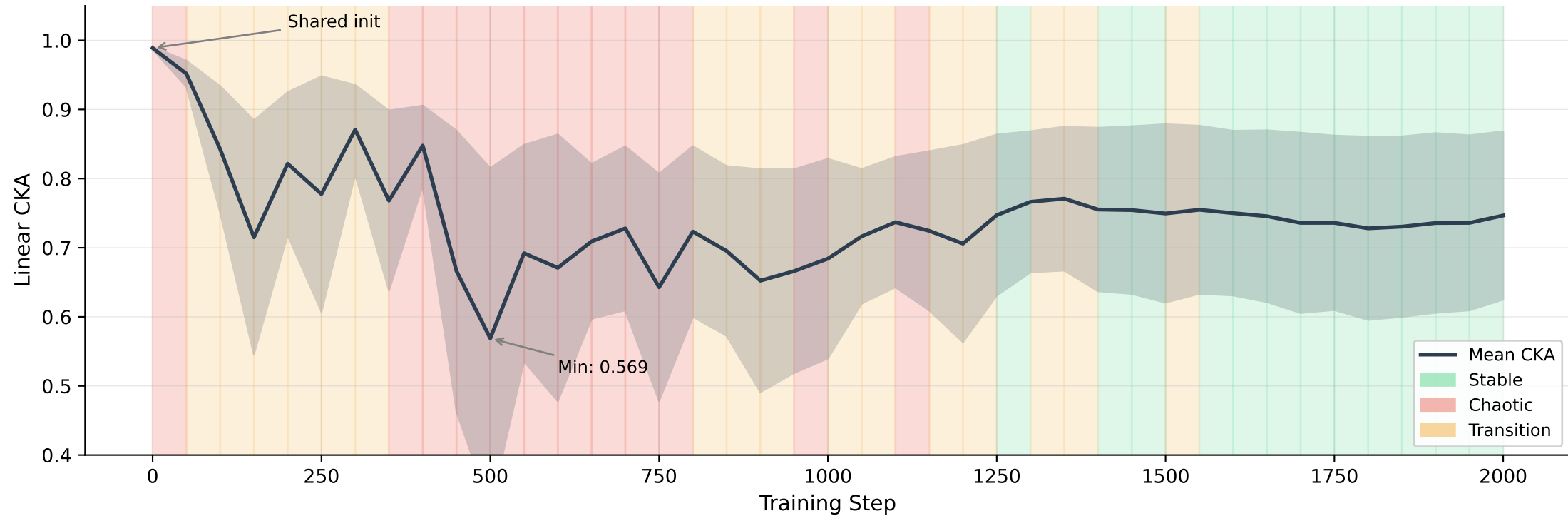
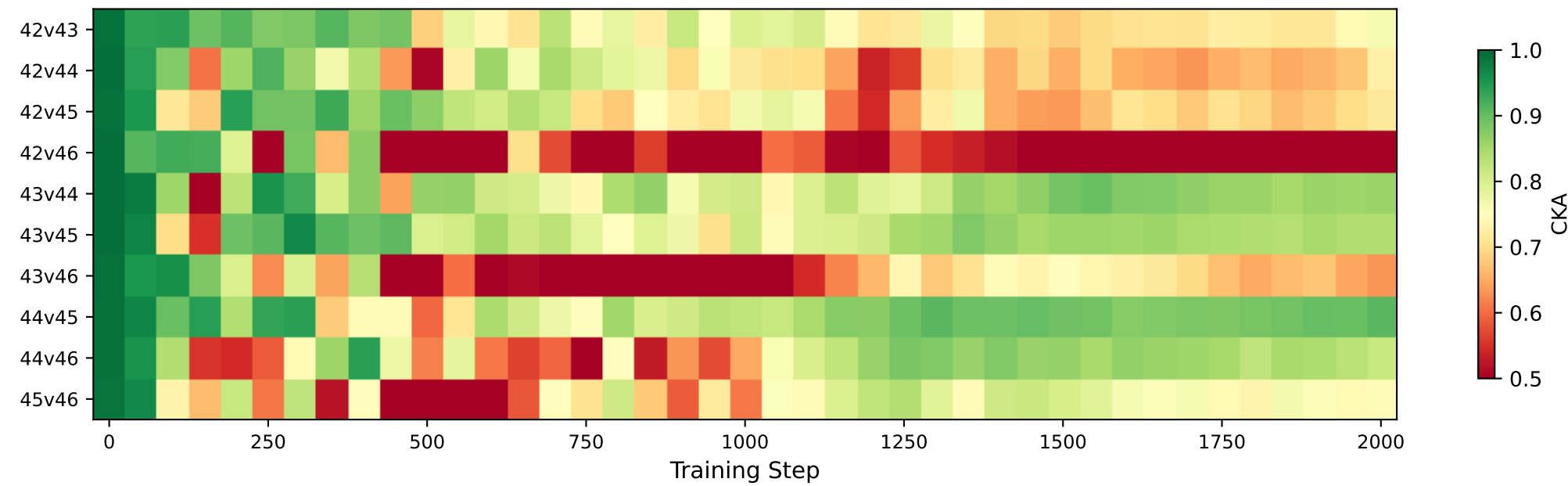


Phase 1: Ensemble Convergence Evidence

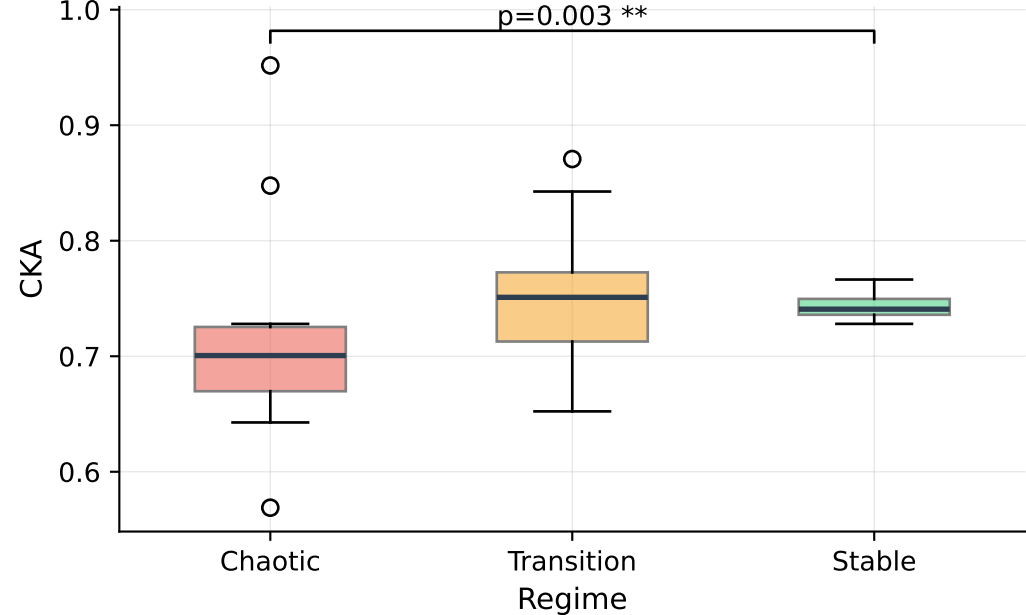
Cross-Seed Representational Alignment (CKA)



Pairwise CKA Across Seed Pairs



CKA by Regime Type



Cross-Seed Representational Analysis

Seeds: [42, 43, 44, 45, 46]
Checkpoints: 41
Seed pairs: 10

CKA trajectory:

Init (shared): 0.989
Min (chaotic): 0.569
Late stable: 0.735

Statistical tests:

Stable vs Chaotic (MW): $p=0.0030$
Late stable vs early chaotic:
 $p=0.0096$

Interpretation:

Models diverge during chaotic regimes then re-converge in stable regimes, supporting ensemble collapse.